FORECASTING CARDIOVASCULAR MEDICATION UTILISATION DURING FLOOD DISASTERS:

A MULTI-HORIZON COMPARISON OF XGBOOST, LSTM AND TEMPORAL FUSION TRANSFORMER

KEMENTERIAN KESIHATAN MALAYSIA

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INTRODUCTION

- Flood disasters can **disrupt** healthcare services and affect medication demand.[1]
- Accurate demand forecasting can **support medication planning** during disaster periods.[2]
- However, evidence on machine learning approaches for forecasting medication utilisation during floods **remains limited**.[2,3]

OBJECTIVE

To compare the forecasting performance of
1) XGBoost,
2) Long Short-Term Memory (LSTM),
3) Temporal Fusion Transformer (TFT) models for predicting cardiovascular medication utilisation across **7-, 14-, 21-, and 28-day** forecast horizons.

METHODS

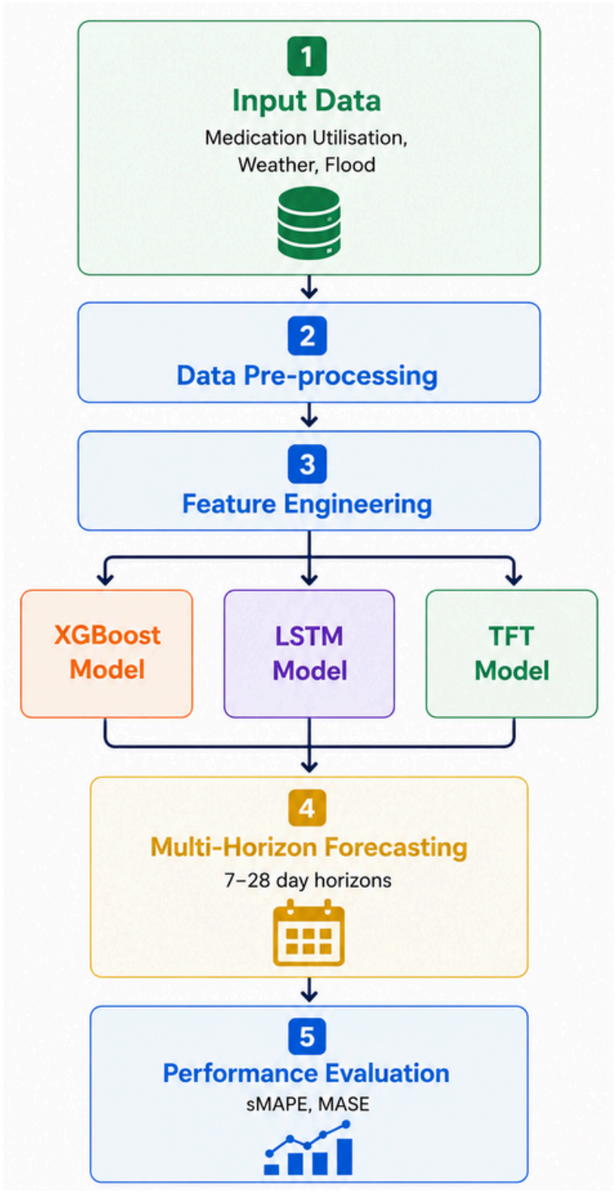


Figure 1. Overview of the proposed framework.

sMAPE: Symmetric Mean Absolute Percentage Error
MASE: Mean Absolute Scaled Error

RESULTS & DISCUSSION

How does forecasting performance change across forecast horizons?

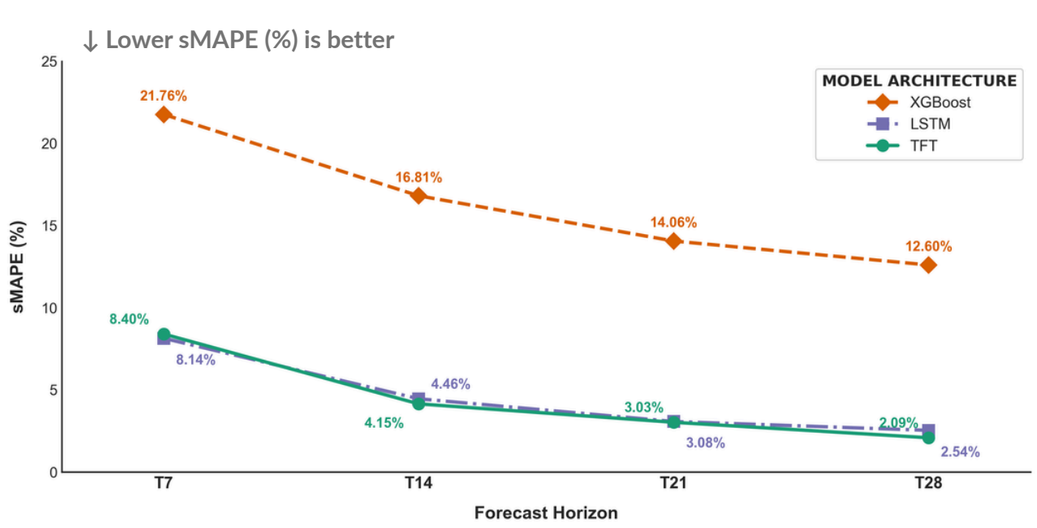


Figure 2. Overall forecasting performance across four horizons.

Prediction error decreased with longer forecast horizons.

This may be partly explained by the rolling-sum target. Longer forecast horizons aggregate medication utilisation over more days and smooth short-term variability. Therefore, lower error at longer forecast horizons does not necessarily mean that longer-term forecasting is easier.

2 How do models perform during flood observations?

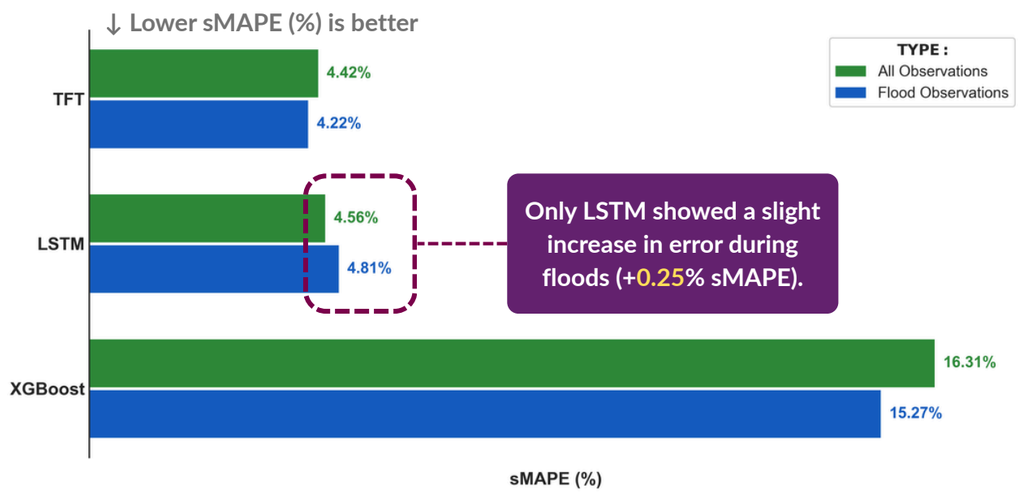


Figure 3. Mean sMAPE across 7- to 28-day horizons.

Model performance remained stable during flood observations, with only LSTM showing a slight increase in average sMAPE.

XGBoost showed higher error and relied heavily on historical utilisation features. LSTM and TFT incorporated a broader range of historical, operational and environmental information, which may have contributed to their stronger performance during flood observations.

3 Which model performs best across the tested settings?

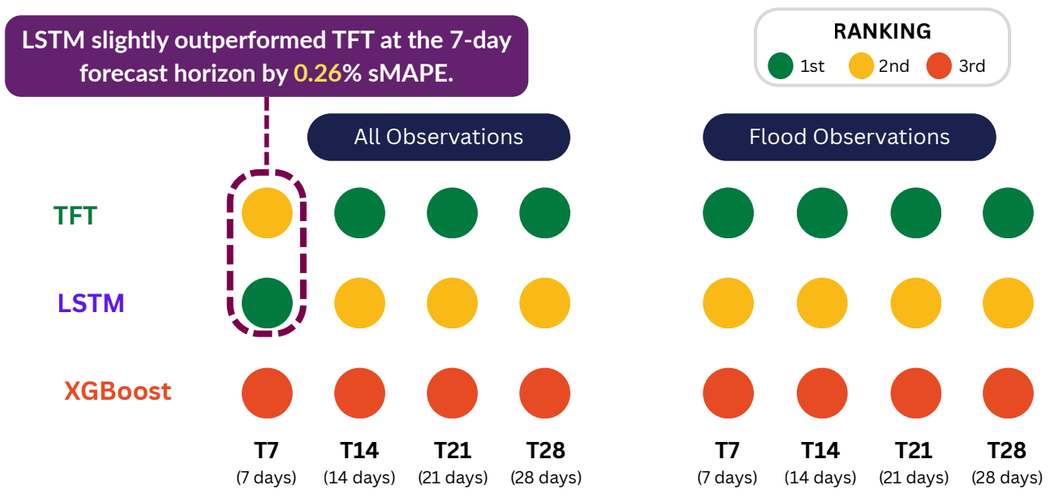


Figure 4. Model ranking across four horizons for overall and flood evaluations.

TFT was the best-performing model in 7 of 8 evaluation settings.

TFT variable selection retained historical utilisation together with temporal and flood-related variables, while temporal attention placed greater weight on recent observations. This may have contributed to TFT performance across forecast horizons.

CONCLUSION

Cardiovascular medication utilisation could be forecast up to **28 days**, with LSTM and TFT outperforming XGBoost. Forecasting error decreased at longer horizons, potentially reflecting smoothing from the rolling-sum targets. **TFT was the best-performing model** in 7 of 8 evaluation settings.

PRACTICAL IMPACT

Provides **quantitative forecasts across multiple horizons** to support advance medication planning and resource preparedness during flood-related healthcare disruptions.

FUTURE WORK

External validation in new settings using an **applicability domain** framework.

MORE INFO

Scan the QR code to explore the study and supplementary information, or visit:
radziaziz.github.io/med-flood



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JABATAN PENGAIRAN DAN SALIRAN MALAYSIA

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